MOCK EXAM — ĐỀ THI MẪU

Satellite Remote Sensing of Atmospheric Composition

Duration: 120 minutes • 10 questions • 25 points • Closed book, calculator allowed

Instructions

- Answer all questions in English. Use clear schematics where asked; label axes and arrows.
- Give the main equations and orders of magnitude; justify every choice in one or two sentences.
- Mỗi câu có ghi chương / phần tương ứng trong đề cương để ôn lại (Part A, Ch. 1–8).

Question 1 (2.5 pts) — see study guide Ch. 1

(a) Define the atmospheric lifetime τ of a pollutant. How are lifetimes combined when several removal processes act in parallel?

(b) The rate constant of CO + OH is $k \approx 2\times10^{-13}$ cm^3 molecule^{-1} s^{-1} and the mean OH concentration is $\approx 1\times10^6$ molecule cm^{-3}. Estimate the lifetime of CO with respect to OH (give the result in days).

(c) A TROPOMI map shows NO_2 concentrated in small hotspots over cities, while a map of CH_4 from the same instrument is almost uniform. Explain the difference.

Question 2 (2.5 pts) — see study guide Ch. 2

(a) Write the Beer–Lambert law for a beam crossing an absorbing and scattering medium. Define optical depth and transmittance.

(b) Explain Kirchhoff's law and why it matters for thermal-infrared remote sensing.

(c) A beam crosses a layer with vertical optical depth $\tau = 0.5$ at a solar zenith angle of 60°. What fraction of the direct beam is transmitted?

Question 3 (2.5 pts) — see study guide Ch. 3

(a) Explain the three main steps of a DOAS retrieval of the tropospheric NO_2 column from a nadir UV–visible spectrometer.

(b) Why does the air mass factor depend on the a priori shape of the vertical profile and on clouds?

(c) Give two physical reasons why a nadir sounder can obtain some information on the vertical distribution of a gas.

Question 4 (2.5 pts) — see study guide Ch. 4

(a) Sketch the nadir, limb and solar occultation geometries.

(b) Which geometry would you choose to (i) monitor long-term trends of stratospheric ozone with the highest accuracy, (ii) map NO_2 emissions from individual cities? Justify.

(c) What does a geostationary orbit add compared to a low-Earth sun-synchronous orbit for air-quality monitoring?

Question 5 (2.5 pts) — see study guide Ch. 5

(a) Write the relation between the retrieved profile, the true profile and the a priori using the averaging kernel matrix.

(b) A retrieval of CO from a thermal-infrared sounder has DOFS = 1.2. What does this mean? Where in the atmosphere is the sensitivity usually highest, and why?

(c) Explain how averaging kernels must be used when comparing satellite retrievals with a chemistry-transport model.

Question 6 (2.5 pts) — see study guide Ch. 6

(a) Define the aerosol optical depth and the Ångström exponent. What does an Ångström exponent of 1.8 tell you about the particles?

(b) Why is it easier to retrieve AOD over the ocean than over a desert in the visible? Which strategy is used over bright surfaces?

(c) Compare MODIS (passive) and CALIOP (lidar) for studying a Saharan dust plume over the Atlantic.

Question 7 (2.5 pts) — see study guide Ch. 7

(a) Compute the size parameter x for (i) an N_2 molecule ($r \approx 0.2$ nm), (ii) a sulfate particle $r = 0.2$ µm, (iii) a dust particle $r = 5$ µm, all at $\lambda = 0.55$ µm. Give the scattering regime of each.

(b) Why is the sky blue but clouds white?

(c) Sketch the phase function in the Rayleigh regime and for a large particle.

Question 8 (2.5 pts) — see study guide Ch. 8

(a) Explain why the O_2 A-band (≈ 760 nm) is suitable for retrieving aerosol or cloud height. Why O_2 and not another gas?

(b) In the thermal infrared, a dust layer at 5 km and another at 1 km have the same AOD. Which one produces the larger brightness-temperature depression, and why? When does the method fail?

(c) Name one other passive technique giving plume heights.

Question 9 (2.5 pts) — see study guide Part A.1–A.4

(a) Write the radiative transfer equation dI/dl = … in the lecture notation. Explain each term and list the processes that add photons to, or remove photons from, the line of sight.

(b) For a layer of gas without scattering in thermodynamic equilibrium, write the integrated radiative transfer equation and explain its two terms. Include the surface boundary term with emissivity ε_s.

(c) A homogeneous layer, $L = 1$ km, contains an absorber with $N = 5\times10^{13}$ molecules cm^{-3} and $\sigma = 2\times10^{-20}$ cm^2. Compute τ and the transmission t.

Question 10 (2.5 pts) — see study guide Part A.5–A.6

(a) Give the physical origin and the dependence on T and P of the Doppler and Lorentz line widths. Which one dominates at 5 km and at 40 km?

(b) Explain why the variation of line widths with altitude provides information on the vertical distribution of an absorber.

(c) Explain the principle of DOAS using $\sigma = \sigma_b + \sigma'$. Why is a polynomial included in the fit? If $\sigma' = 2\times10^{-19}$ cm^2 and the slant column is 1×10^{16} molecules cm^{-2}, what is the differential optical depth?

Suggested answer key — Đáp án gợi ý

Short outlines only: a full answer develops each point with the equations and figures of the corresponding chapter of the study guide.

Question 1 (Ch. 1)

- **(a)** τ = burden / loss rate (M / L); for first-order sinks $\tau = 1/k'$. Parallel sinks: $1/\tau = \Sigma\, 1/\tau_i$ (chemistry, photolysis, dry and wet deposition).
- **(b)** $\tau = 1/(k[OH]) = 1/(2\times10^{-13} \times 1\times10^{6}) = 5\times10^{6}$ s ≈ **58 days ≈ 2 months**.
- **(c)** NO_2 lives hours → destroyed before transport spreads it → map = sources. CH_4 lives ~10 years ≫ global mixing time (~1 yr) → well mixed, variations of ~1 %.

Question 2 (Ch. 2)

- **(a)** dI = −k_e I ds, $I = I_0$ e^(−τ), $\tau = \int(k_a + k_s)\, ds = \int \Sigma\sigma_i n_i\, ds$; transmittance $T = I/I_0 = e^{(-\tau)}$.
- **(b)** In LTE, emissivity = absorptivity at each wavelength (ε_λ = α_λ): a gas that absorbs at λ also emits B_λ(T) there → TIR spectra carry information on both concentration and temperature.
- **(c)** Slant τ = 0.5 / cos 60° = 1.0 → T = e^(−1) ≈ **0.37 (37 %)**.

Question 3 (Ch. 3)

- **(a)** 1) Spectral fit of $\ln(I_0/I)$ with lab cross-sections + polynomial → slant column SCD. 2) Remove stratospheric part. 3) Divide by tropospheric AMF (RT model) → VCD.
- **(b)** AMF = SCD/VCD measures the light-path length; sensitivity varies with altitude (low near surface, shielded below cloud, enhanced above) → the result depends on where the gas is assumed to be.
- **(c)** Absorption strength varies with λ (different penetration depths); pressure broadening of lines (TIR); temperature contrast; (limb geometry for limb sounders).

Question 4 (Ch. 4)

- **(a)** See Figure 'Nadir, limb and occultation viewing geometries' in the study guide.
- **(b)** (i) Solar occultation: self-calibrating (I/I_0), high vertical resolution → trend quality. (ii) Nadir: km-scale horizontal resolution, daily global coverage, sensitive to the boundary layer.
- **(c)** Hourly revisit over one continent → diurnal cycle of pollution (rush hours, photochemistry), more cloud-free scenes. LEO: one overpass per day at fixed local time, but global.

Question 5 (Ch. 5)

- **(a)** $\hat{x} = x_a + A(x_{true} - x_a) + G\varepsilon$; $A = \partial\hat{x}/\partial x = GK$; DOFS = trace(A).
- **(b)** ≈ one independent piece of information → essentially a column, weighted towards the mid-troposphere (≈ 3–8 km) where thermal contrast with the surface is largest; little near the surface.
- **(c)** Smooth the model profile with the AKs: $x_{m,smoothed} = x_a + A(x_{model} - x_a)$, then compare with $\hat{x}$ (removes the smoothing and a-priori effects).

Question 6 (Ch. 6)

- **(a)** AOD = $\int k_{ext,aer}\, dz$; $\tau(\lambda) \propto \lambda^{(-\alpha)}$. α ≈ 1.8 → fine particles (smoke, urban/sulfate).
- **(b)** Dark ocean → aerosol reflectance dominates TOA signal; bright desert masks the aerosol contribution. Use deep blue (412 nm, dark surfaces), multi-angle/polarisation, or TIR.
- **(c)** MODIS: wide swath, daily AOD map, size info, but no altitude, daytime only. CALIOP: vertical profile, day+night, depolarisation (non-spherical dust), but narrow curtain, 16-day revisit.

Question 7 (Ch. 7)

- **(a)** $x = 2\pi r/\lambda$: (i) ≈ 0.002 → Rayleigh; (ii) ≈ 2.3 → Mie; (iii) ≈ 57 → upper Mie / geometric optics limit.

- **(b)** Molecules: $\sigma \propto \lambda^{-4}$ → blue scattered much more. Cloud droplets ($x \gg 1$): $Q_{ext} \approx 2$, independent of λ → all colours scattered equally → white.
- **(c)** Rayleigh: $\frac{3}{4}(1 + \cos^2\Theta)$, symmetric forward/back. Large particle: strong forward peak, weak side/back lobes.

Question 8 (Ch. 8)

- **(a)** O_2 is well mixed with a known profile (follows pressure); a higher scattering layer shortens the photon path → weaker O_2 absorption. The depth/shape of the band ↔ layer pressure.
- **(b)** The 5 km layer: colder (≈ 6.5 K/km lapse rate) → larger thermal contrast with the surface → larger ΔBT. Fails with low thermal contrast (low layers, night inversions, cold surfaces).
- **(c)** Stereo/multi-angle parallax (MISR); also UV aerosol index dependence on height, O_2–O_2 band at 477 nm.

Question 9 (Part A.1–A.4)

- **(a)** $dI(\tilde{\nu})/dl = -k_{ext} \cdot I + J$; $k_{ext} = \alpha_{gas} + s_{gas} + \alpha_{aer} + s_{aer}$ (extinction); J = source (emission, scattering into the line of sight). Gains: surface reflection, single/multiple scattering into the LOS, emission. Losses: absorption, scattering out of the field of view.
- **(b)** $J = \alpha \cdot B(T)$; $I(L) = I(0)e^{(-\tau(0,L))} + \int B(T(l))\, e^{(-\tau(l,L))}\, d\tau$: attenuated background + emission of each layer attenuated by the gas above. $I(0) = \varepsilon_s B(T_s) + (1 - \varepsilon_s) I_s$, $1 - \varepsilon_s$ = surface albedo.
- **(c)** $\tau = \sigma \cdot N \cdot L = 2\times10^{-20} \times 5\times10^{13} \times 10^{5} =$ **0.1** → $t = e^{(-0.1)} \approx$ **0.90 (90 %)**.

Question 10 (Part A.5–A.6)

- **(a)** Doppler: thermal motion, Gaussian, $\propto \tilde{\nu}\sqrt{(T/m)}$, independent of P. Lorentz: collisions, $\propto P(T_0/T)^n$. Lorentz dominates at 5 km (high P), Doppler at 40 km (P ≈ 3 hPa).
- **(b)** P(z) is known and decreases exponentially → line shape differs for each altitude: centre formed by high layers, wings by low layers; fitting the shape (with known P, T profiles) gives the profile.
- **(c)** Fast structures σ' = gas fingerprint, slow part σ_b mixed with Rayleigh/Mie/surface → removed by the polynomial. $D = \sigma' \cdot SCD = 2\times10^{-19} \times 10^{16} =$ **2×10^{-3}** (0.2 % signal).

Grading hints — Cách chấm thường gặp

- Definition + equation + order of magnitude ≈ 50 % of the points of each sub-question.
- A correct, labelled schematic is often worth as much as a paragraph of text.
- Always compare advantages AND disadvantages when asked to choose a method or geometry.